

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I _G.Jaahnvi(192525277)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: _____G.jaahanvi_____

Assessment Tasks Class Test

1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.
2. Explain the 5Vs of Big Data with one practical example for any two Vs.
3. Differentiate structured, semi-structured, and unstructured data with examples.
4. Explain the role of distributed file systems in big data processing.
5. List any three major security concerns in big data platforms and briefly explain each.
6. What are data appliances and integration tools? Explain their role in big data workflows.
7. Discuss two major challenges organizations face while implementing big data analytics.
8. Write short notes on any one domain application of big data analytics.
9. How do business drivers influence the adoption of big data solutions?
10. Explain the importance of big data analytics in decision making.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage

Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Define Big Data and explain why traditional data processing approaches are often insufficient.

Big Data refers to extremely large, complex, and rapidly generated datasets that cannot be efficiently stored, processed, or analyzed using traditional data-processing systems.

Characteristics of Big Data

Big Data is commonly described using the **5Vs**:

- **Volume** – huge amounts of data.
- **Velocity** – data is generated and processed very quickly.
- **Variety** – different types and formats of data.
- **Veracity** – accuracy and reliability of data.
- **Value** – useful information obtained from data.

Why traditional approaches are insufficient

Traditional databases and processing systems face several limitations:

1. **Large data volume:** Traditional systems may not efficiently handle terabytes or petabytes of data.
2. **High processing requirements:** Processing huge datasets using a single computer can be very slow.
3. **Different data formats:** Big Data includes text, images, videos, sensor data, logs, and social-media data.
4. **High data generation speed:** Traditional systems may not process continuously generated data quickly enough.
5. **Scalability issues:** Increasing the capacity of a single machine can become expensive.
6. **Fault tolerance:** Failure of one system can affect the entire processing operation.

Therefore, Big Data technologies use **distributed storage and parallel processing**, where data and computation are distributed across multiple machines.

2. Explain the 5Vs of Big Data with practical examples.

The 5Vs describe the major characteristics of Big Data.

1. Volume

Volume refers to the huge quantity of data generated and stored.

Example:

Social-media platforms generate enormous amounts of posts, images, videos, and messages every day.

2. Velocity

Velocity refers to the speed at which data is generated, transmitted, and processed.

Example:

Stock-market systems receive and process thousands of transactions within seconds.

3. Variety

Variety refers to the different forms of data.

Examples include:

- Structured data – tables and databases.
- Semi-structured data – JSON and XML.
- Unstructured data – images, videos, audio, and documents.

4. Veracity

Veracity represents the quality, accuracy, and reliability of data.

Example:

Customer reviews may contain incorrect information, duplicate entries, or fake reviews. Analytics systems need to identify and handle such unreliable data.

5. Value

Value refers to the useful information and benefits obtained from data.

Example:

A shopping website analyzes customer purchases to recommend products that customers may be interested in.

3. Differentiate structured, semi-structured, and unstructured data.

Feature	Structured Data	Semi-structured Data	Unstructured Data
Meaning	Highly organized data	Partially organized data	Data without a fixed structure
Format	Rows and columns	Tags, keys, metadata	No predefined format
Schema	Fixed schema	Flexible schema	No fixed schema
Storage	Relational databases	NoSQL/document databases	Distributed storage/cloud storage
Examples	Student database	JSON, XML	Images, videos, audio
Processing	Relatively easy	Moderately complex	More complex

4. Explain the role of distributed file systems in Big Data processing.

A **Distributed File System (DFS)** stores large datasets across multiple computers or nodes instead of storing everything on a single machine.

Role of DFS

- Distributed storage:**
Large files are divided into smaller blocks and stored across multiple machines.
- Scalability:**
Additional machines can be added when the amount of data increases.
- Fault tolerance:**
Copies or replicas of data can be maintained. If one machine fails, another copy can be used.
- Parallel processing:**
Multiple machines can process different portions of data simultaneously.
- Efficient data access:**
Data can be accessed from multiple nodes, improving processing performance.
- Handling large datasets:**
DFS is suitable for storing very large datasets that cannot efficiently fit on a single computer.

Example

Hadoop Distributed File System (HDFS) divides large files into blocks and distributes them across different nodes in a Hadoop cluster.

5. List any three major security concerns in Big Data platforms.

1. Data Privacy

Big Data systems store large amounts of personal and sensitive information.

Risk: Unauthorized users may access customer, financial, or personal data.

Solution: Encryption, access control, authentication, and data masking can be used.

2. Unauthorized Access

Attackers or unauthorized employees may gain access to Big Data systems.

Risk: They may modify, steal, or delete important information.

Solution: Strong passwords, multi-factor authentication, role-based access control, and authorization mechanisms should be implemented.

3. Data Breaches

A data breach occurs when confidential information is accessed or exposed without authorization.

Risk: It can cause financial loss, privacy violations, and damage to an organization's reputation.

Solution: Organizations should use encryption, monitoring, security audits, firewalls, and intrusion-detection mechanisms.

6. What are data appliances and integration tools? Explain their role.

Data appliances are specialized systems that combine computing, storage, and data-processing capabilities for efficient data management.

Integration tools help collect, transfer, clean, transform, and combine data from different sources.

Their role in Big Data workflows includes:

- Collecting data from different sources.
- Converting data into suitable formats.
- Cleaning and transforming data.

Workflow:

Data Sources → Integration → Storage → Processing → Analytics → Decision Making

7. Discuss two major challenges in implementing Big Data Analytics.

1. Security and Privacy:

Organizations handle large amounts of sensitive data. Protecting this data from unauthorized access and breaches is a major challenge. Encryption, authentication, and access control are required.

2. Data Quality and Management:

Big Data may contain missing, duplicate, incorrect, or inconsistent information. Poor-quality data can produce inaccurate results. Data cleaning and validation are therefore necessary.

Other challenges include **high cost, lack of skilled professionals, and complex data integration.**

8. Write short notes on one domain application of Big Data Analytics.

Big Data Analytics in Healthcare

Healthcare is an important application area of Big Data Analytics.

Hospitals and healthcare organizations generate large amounts of data from:

- Electronic health records
- Medical images
- Laboratory reports
- Wearable devices

Applications

1. Disease prediction:

Historical patient data can be analyzed to identify disease patterns.

2. Personalized treatment:

Patient information can help doctors select appropriate treatment strategies.

3. Patient monitoring:

Data from monitoring devices can help track patient conditions.

4. Hospital management:

Analytics can help manage beds, staff, equipment, and resources.

Example

Data collected from wearable devices can be analyzed to monitor health-related measurements and identify unusual patterns.

9. How do business drivers influence the adoption of Big Data solutions?

Business drivers are the organizational needs and goals that encourage companies to adopt Big Data technologies.

Major business drivers

1. **Better decision making**
Organizations use analytics to make decisions based on data rather than assumptions.
2. **Customer understanding**
Customer behavior can be analyzed to understand preferences and requirements.
3. **Competitive advantage**
Companies can use data insights to develop better products and services.
4. **Cost reduction**
Analytics can identify unnecessary expenses and improve resource utilization.

10. Explain the importance of Big Data Analytics in decision making.

Big Data Analytics is the process of examining large and complex datasets to discover useful patterns, relationships, and insights.

Importance in decision making

1. **Data-driven decisions**
Organizations can make decisions using actual data rather than assumptions.
2. **Identifying trends**
Analytics can reveal changing customer, market, and operational trends.
3. **Predictive analysis**
Historical data can be used to predict possible future events.
4. **Improved customer experience**
Customer behavior can be analyzed to provide personalized services.
5. **Risk identification**
Analytics can help identify unusual patterns and potential risks.

Example

A retail company can analyze sales data to determine which products are popular, predict future demand, and plan inventory accordingly.